

Supplementary Material

Before Scientific Emergence Becomes Visible: Independent Evidence for Structural Precursors in Large-Scale Scholarly Knowledge

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Supplementary scope

This supplement preserves the complete software output associated with the frozen PBSK v4.0.0 run. Every software-generated figure is reproduced below. Every software-generated CSV table is included in full as a machine-readable file under `supplementary_data/results_tables/`; the PDF provides an inventory and a readable preview for each table. Empty tables are retained and explicitly identified rather than silently omitted. The supplement therefore separates human-readable presentation from the exact machine-readable computational record.

Table 1: Inventory of all software-generated result tables supplied with this submission.

ID	Software table	Rows	Columns
S1	table_ablation_openalex.csv	9	7
S2	table_book_operationalization.csv	40	7
S3	table_creative_universe_generalization.csv	33	6
S4	table_cross_source_agreement.csv	8	5
S5	table_figure_generation_qc.csv	14	3
S6	table_hypothesis_tests_openalex.csv	69	12
S7	table_incremental_value_openalex.csv	4	8
S8	table_landmark_benchmark.csv	12	8
S9	table_matched_controls.csv	24	6
S10	table_oof_predictions_openalex.csv	1012	7
S11	table_openalex_bookwide_features.csv	108	50
S12	table_power_validation_ablation.csv	9	7
S13	table_power_validation_controls.csv	240	17
S14	table_power_validation_cross_source_agreement.csv	8	5
S15	table_power_validation_features_openalex.csv	1080	50
S16	table_power_validation_features_pubmed.csv	354	50
S17	table_power_validation_hypotheses.csv	69	12
S18	table_power_validation_incremental_value.csv	4	8
S19	table_power_validation_oof_predictions.csv	11880	7
S20	table_power_validation_positive_topics.csv	120	16
S21	table_power_validation_predictive_performance.csv	11	14

Continued on next page

Table 1 (continued)

ID	Software table	Rows	Columns
S22	table_power_validation_pubmed_replication.csv	0	0
S23	table_power_validation_temporal_performance.csv	15	15
S24	table_predictive_performance_openalex.csv	11	14
S25	table_pubmed_bookwide_features.csv	48	50
S26	table_pubmed_replication_hypotheses.csv	0	0
S27	table_pubmed_replication_transformation.csv	5	9
S28	table_region_features_all_sources.csv	1590	50
S29	table_retrieval_qc.csv	1068	12
S30	table_sequence_architecture.csv	6	5
S31	table_temporal_leakage_audit.csv	1590	6
S32	table_temporal_predictive_performance.csv	15	15
S33	table_transformation_outcomes.csv	414	15
S34	table_transformation_tests_openalex.csv	5	9

All software-generated figures

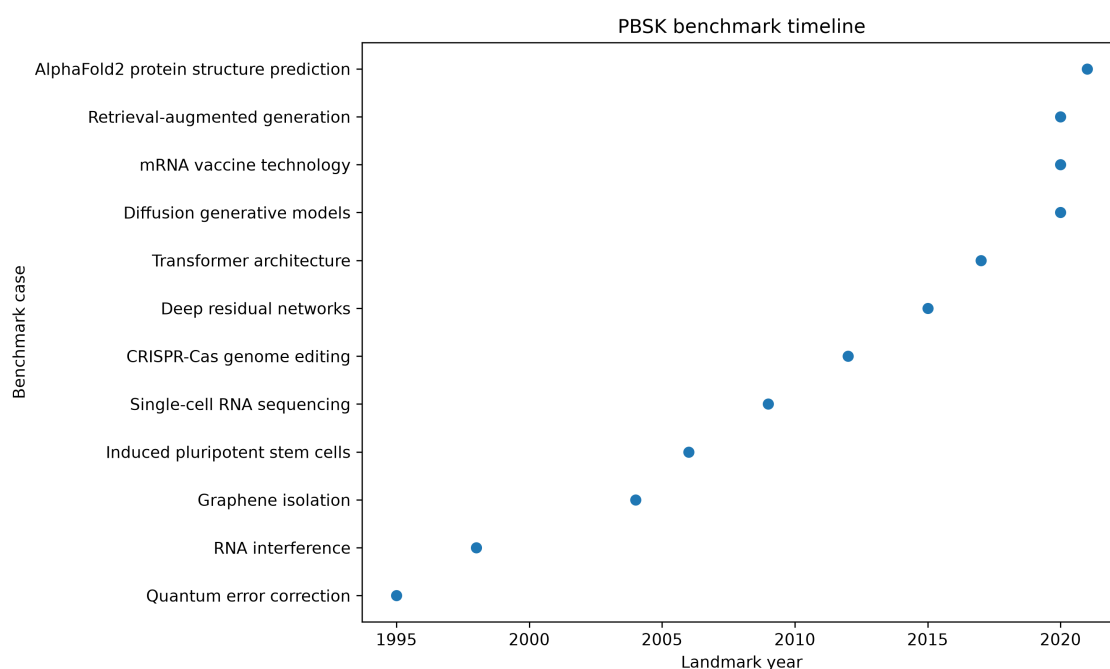


Figure 1: Software-generated figure: figure 01 benchmark timeline. This figure is reproduced without numerical alteration from the frozen PBSK v4.0.0 run.

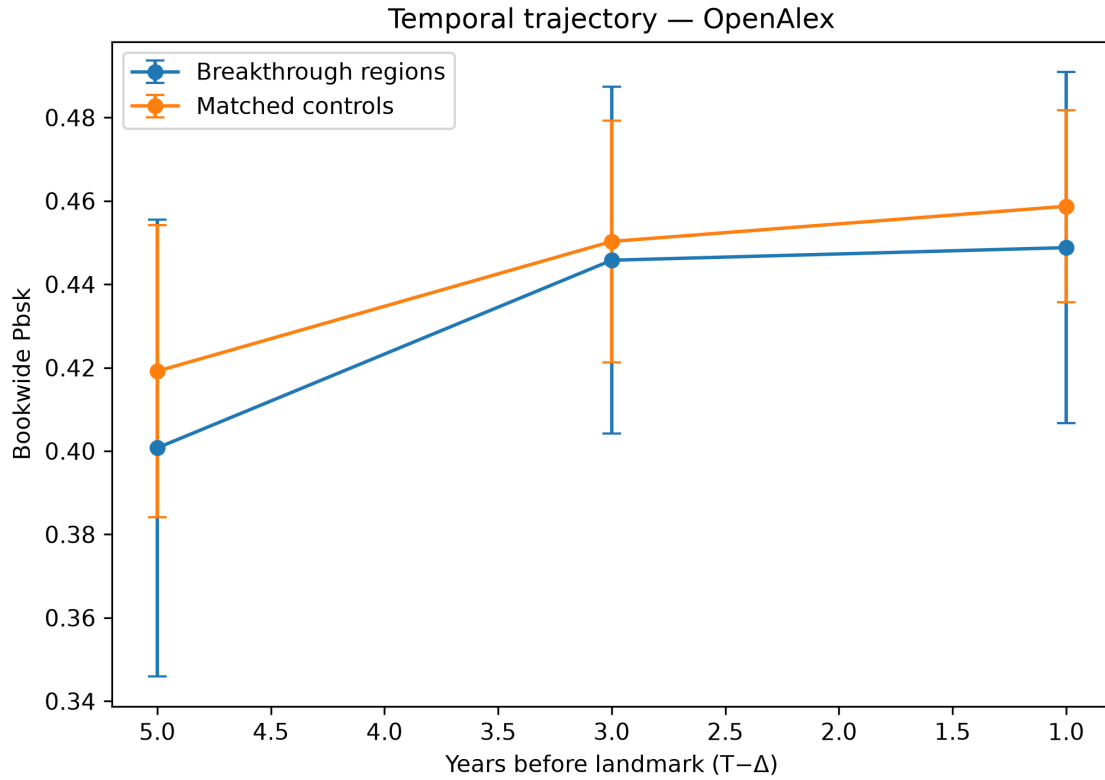


Figure 2: Software-generated figure: figure 02 bookwide pbsk trajectory. This figure is reproduced without numerical alteration from the frozen PBSK v4.0.0 run.

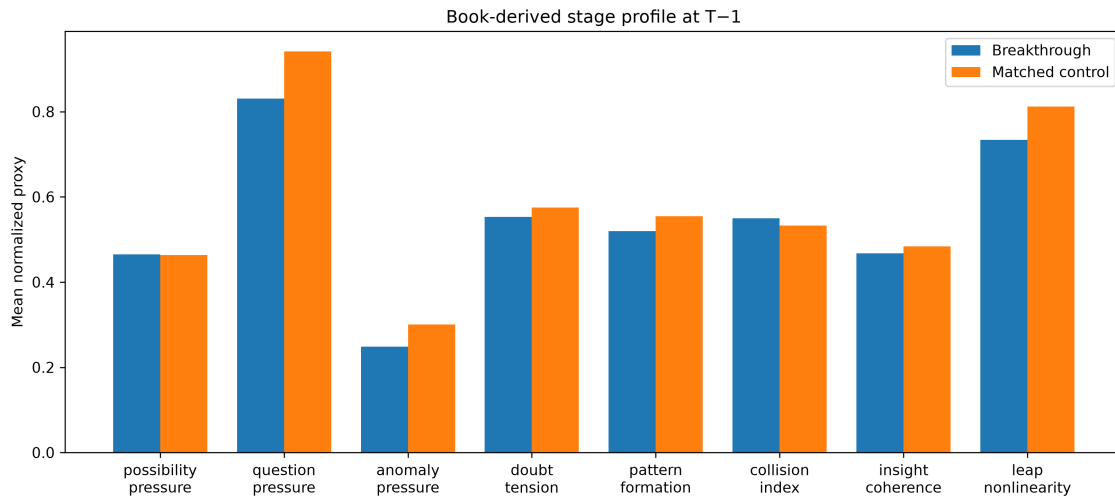


Figure 3: Software-generated figure: figure 03 book stage profile tminus1. This figure is reproduced without numerical alteration from the frozen PBSK v4.0.0 run.

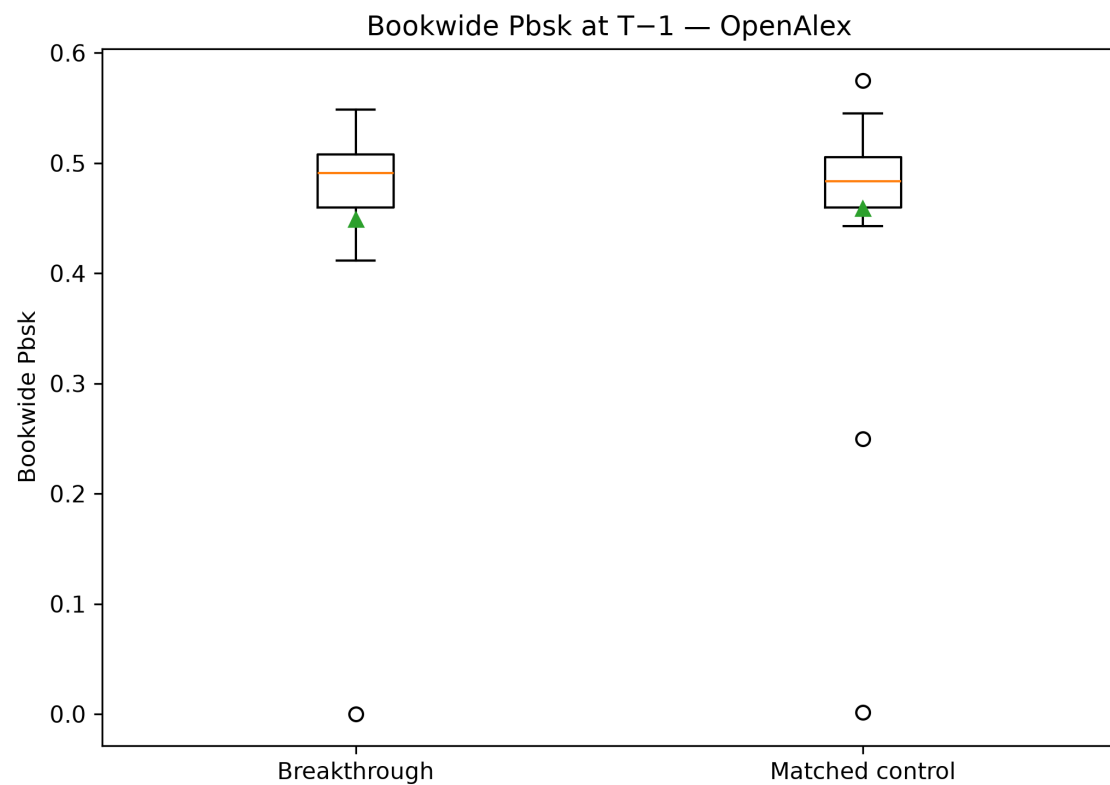


Figure 4: Software-generated figure: figure 04 bookwide pbsk distribution. This figure is reproduced without numerical alteration from the frozen PBSK v4.0.0 run.

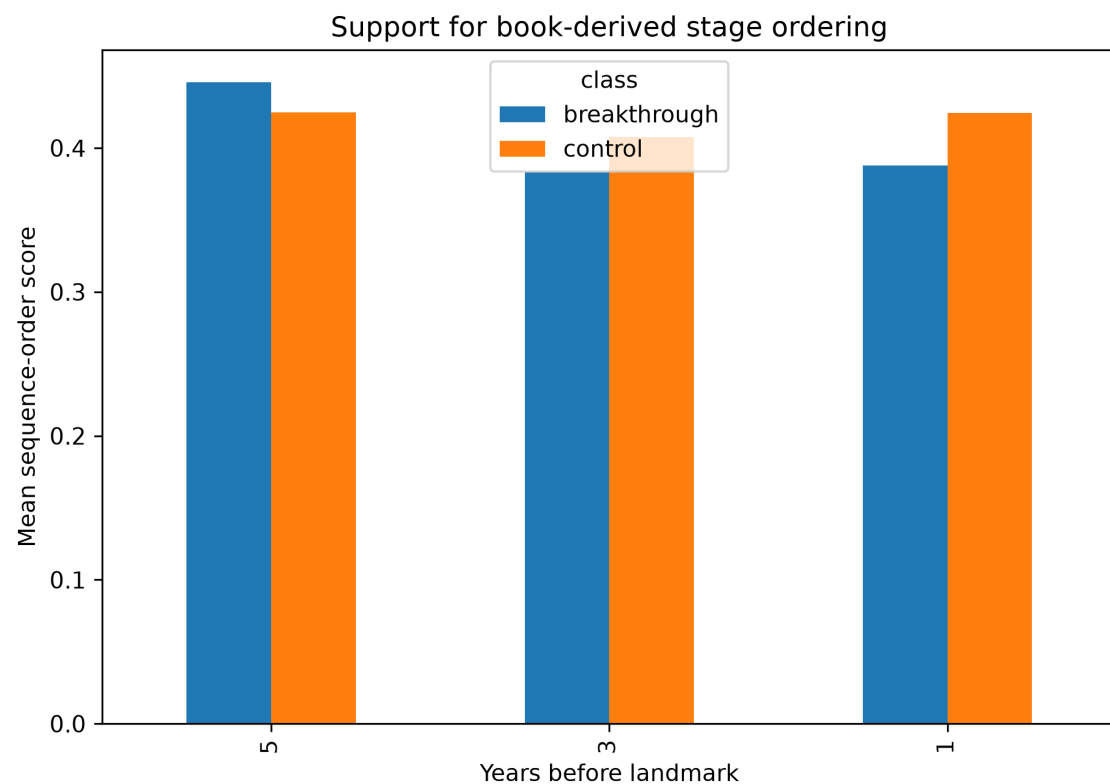


Figure 5: Software-generated figure: figure 05 sequence architecture. This figure is reproduced without numerical alteration from the frozen PBSK v4.0.0 run.

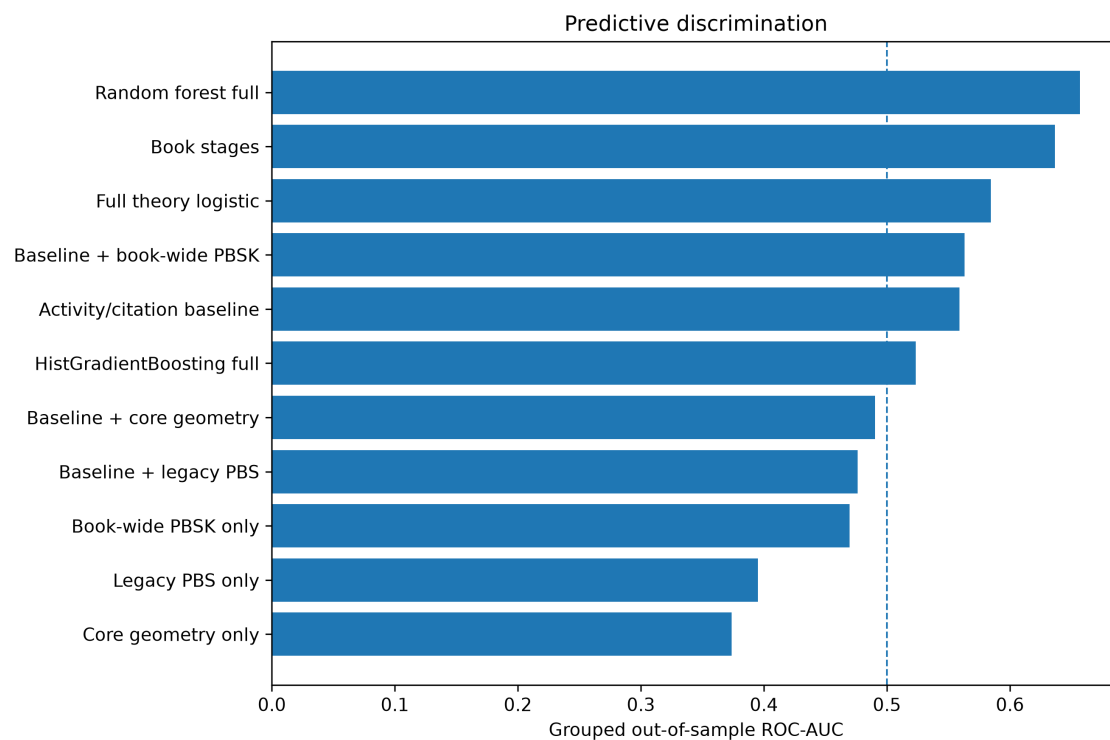


Figure 6: Software-generated figure: figure 06 predictive performance. This figure is reproduced without numerical alteration from the frozen PBSK v4.0.0 run.

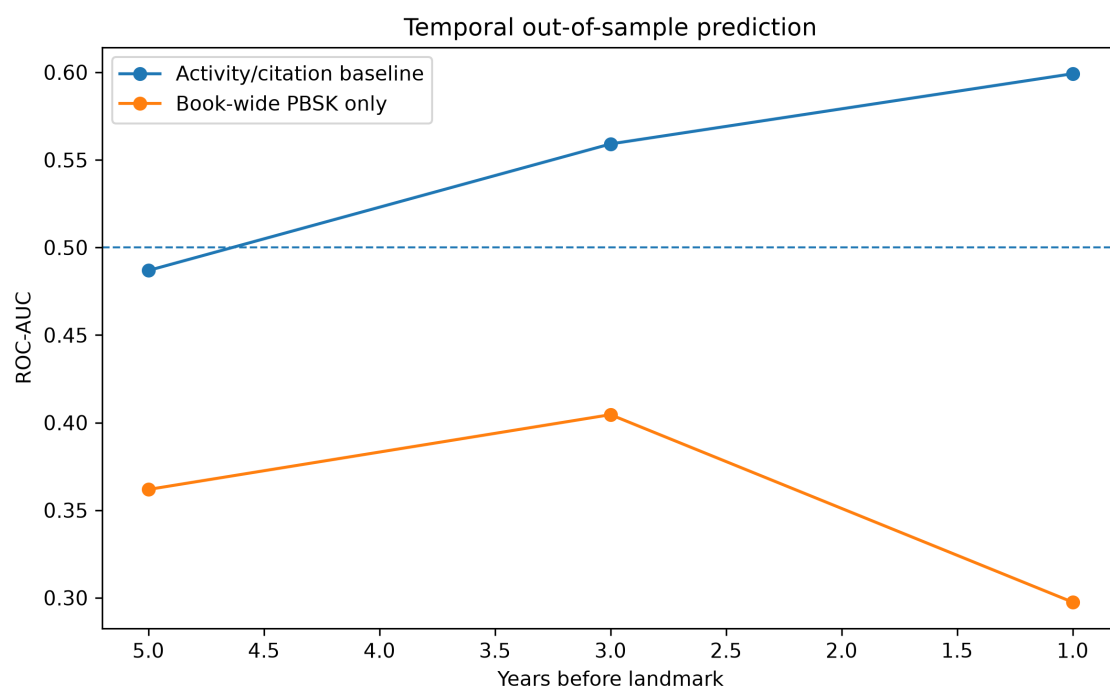


Figure 7: Software-generated figure: figure 07 temporal prediction. This figure is reproduced without numerical alteration from the frozen PBSK v4.0.0 run.

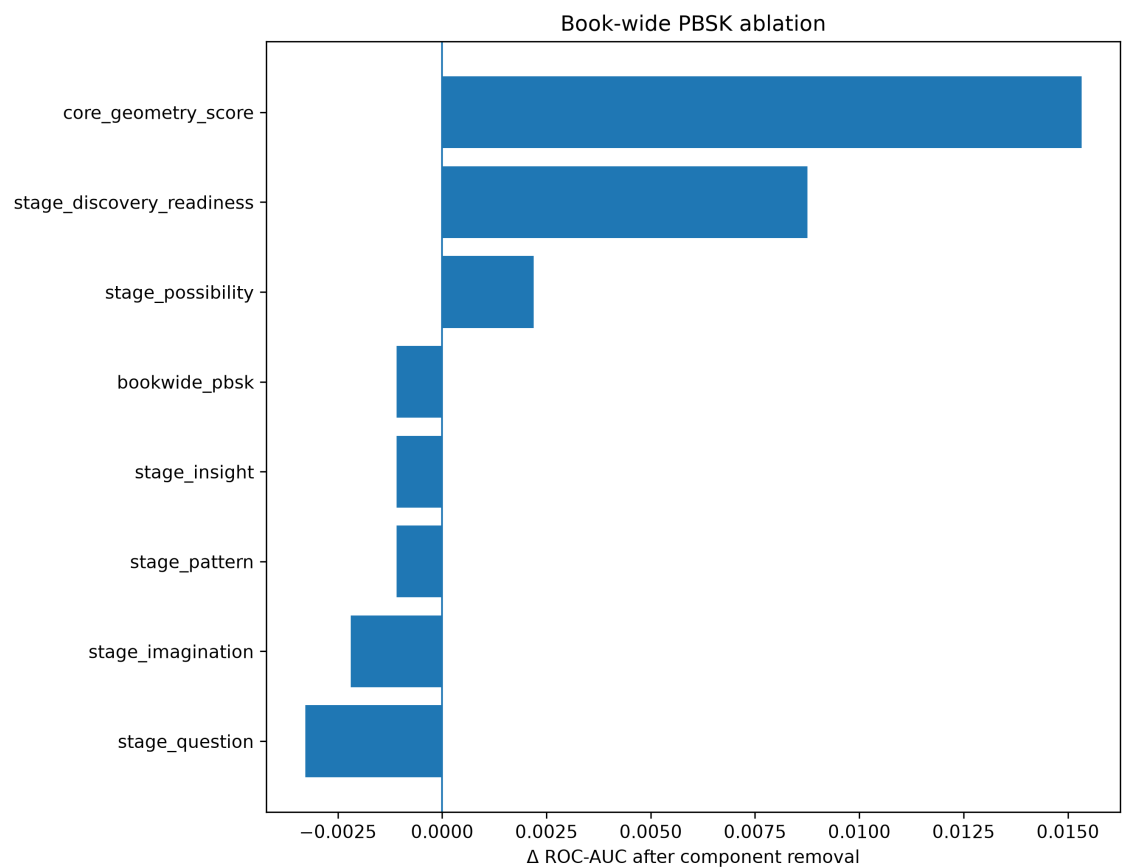


Figure 8: Software-generated figure: figure 08 bookwide ablation. This figure is reproduced without numerical alteration from the frozen PBSK v4.0.0 run.

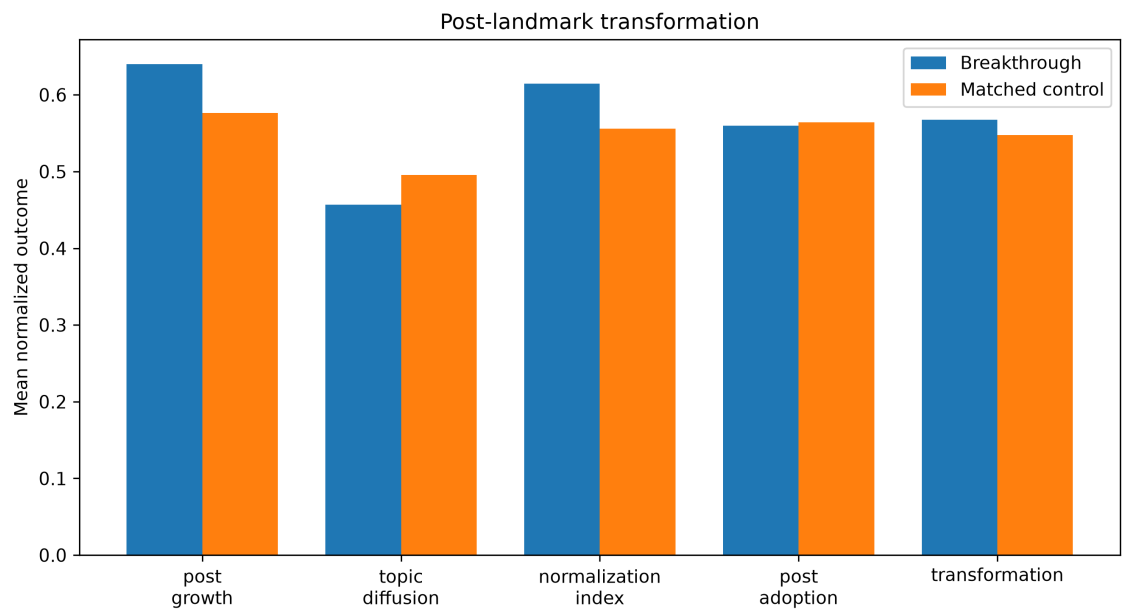


Figure 9: Software-generated figure: figure 09 transformation. This figure is reproduced without numerical alteration from the frozen PBSK v4.0.0 run.

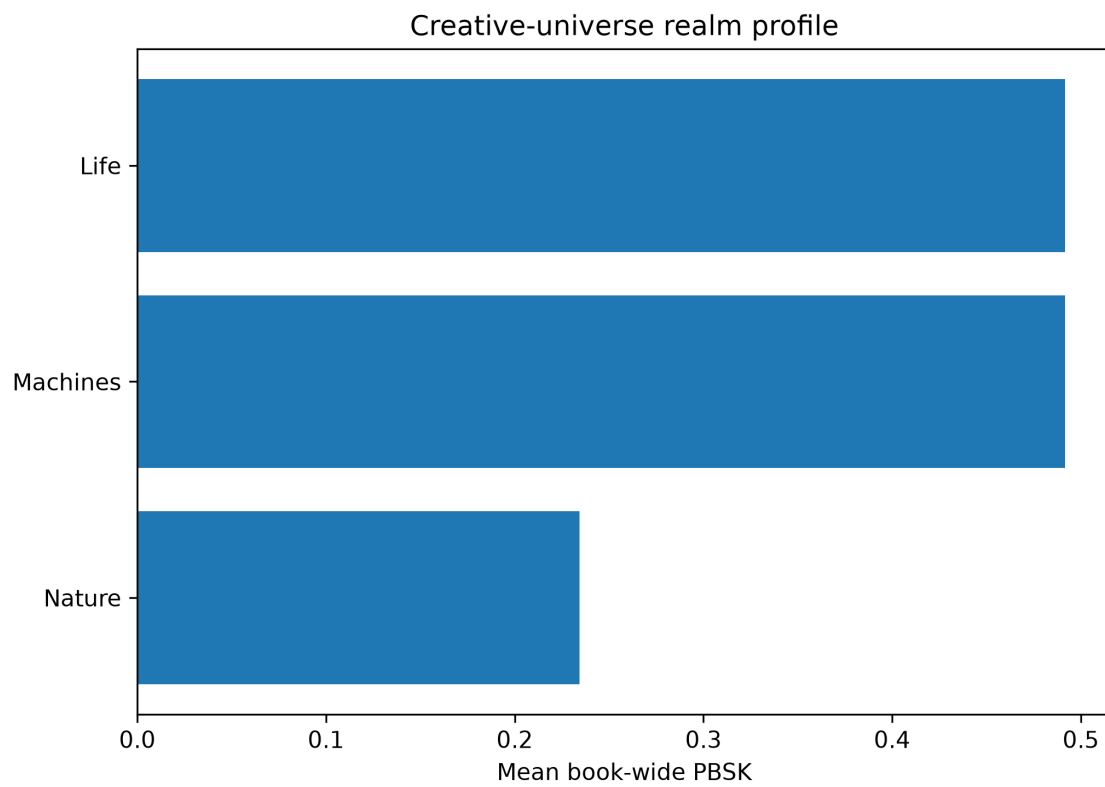


Figure 10: Software-generated figure: figure 10 creative universe realms. This figure is reproduced without numerical alteration from the frozen PBSK v4.0.0 run.

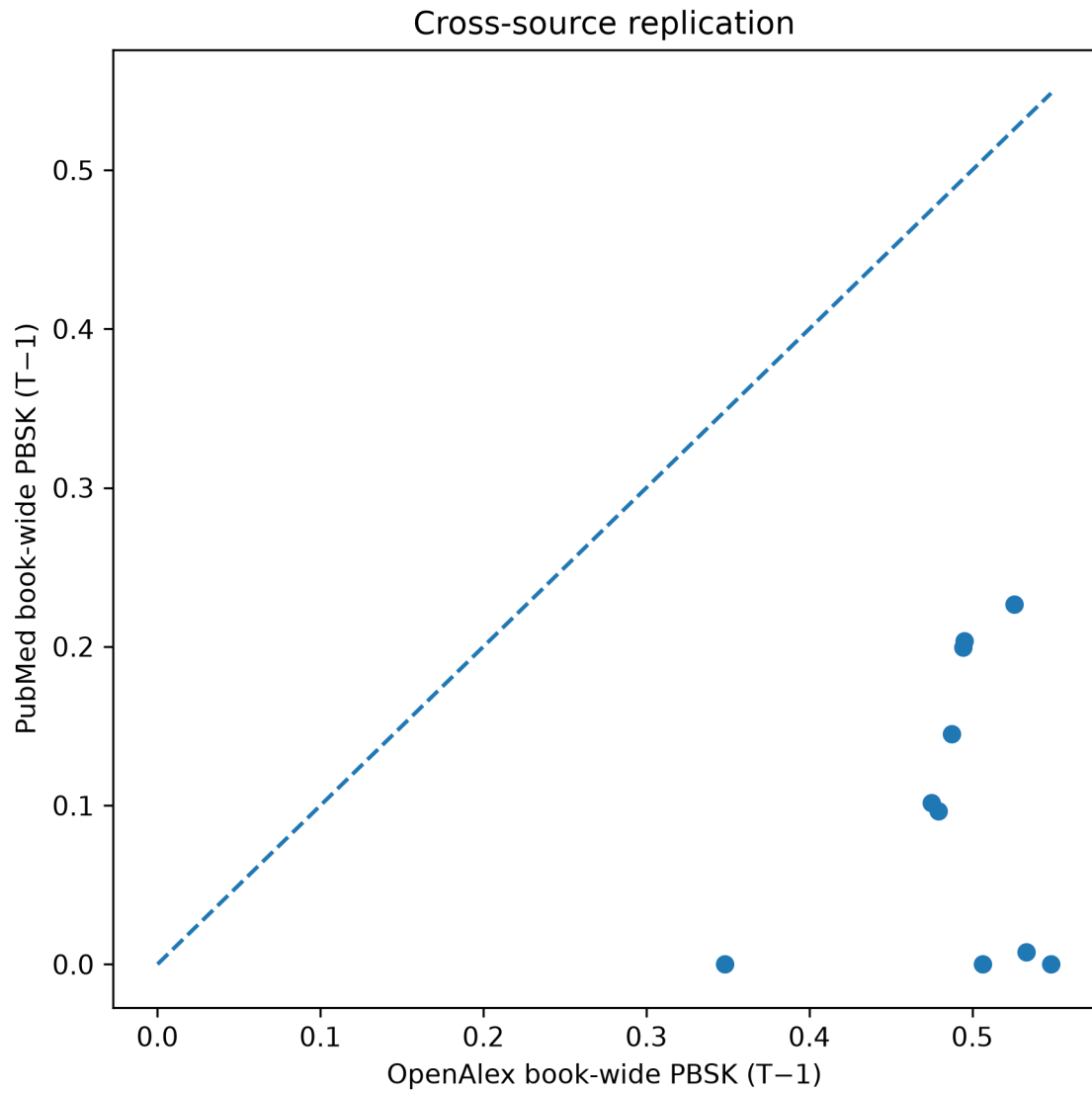


Figure 11: Software-generated figure: figure 11 pubmed replication. This figure is reproduced without numerical alteration from the frozen PBSK v4.0.0 run.

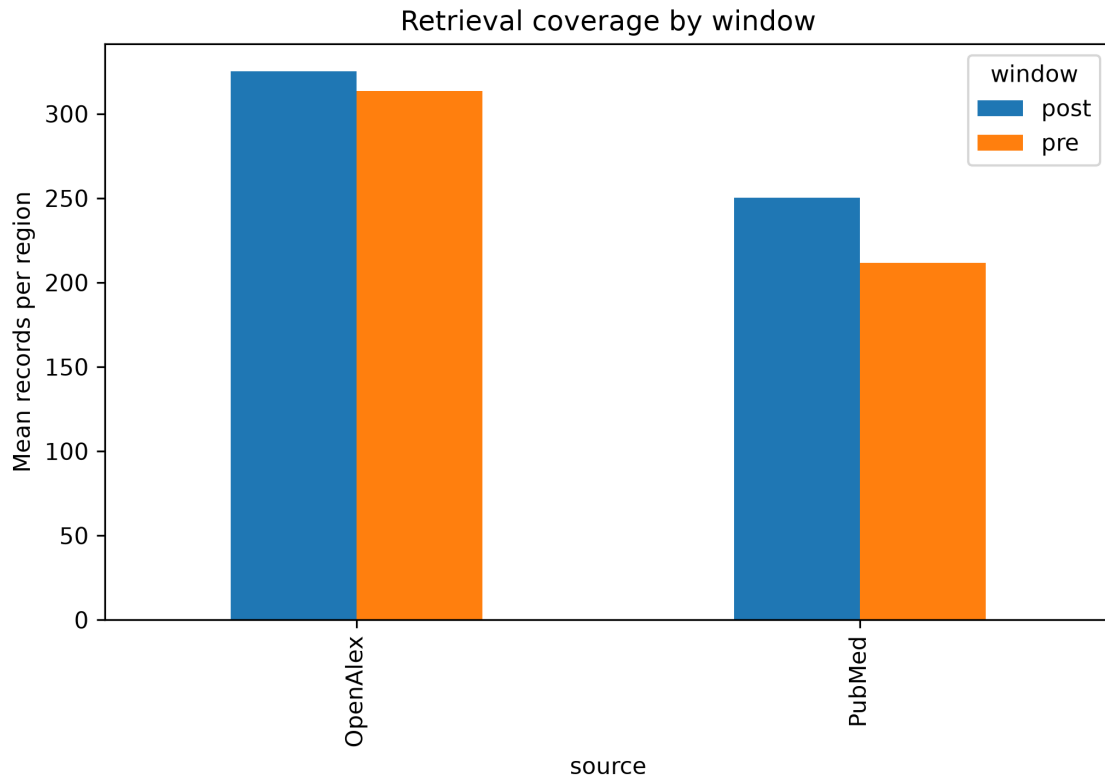


Figure 12: Software-generated figure: figure 12 retrieval qc. This figure is reproduced without numerical alteration from the frozen PBSK v4.0.0 run.

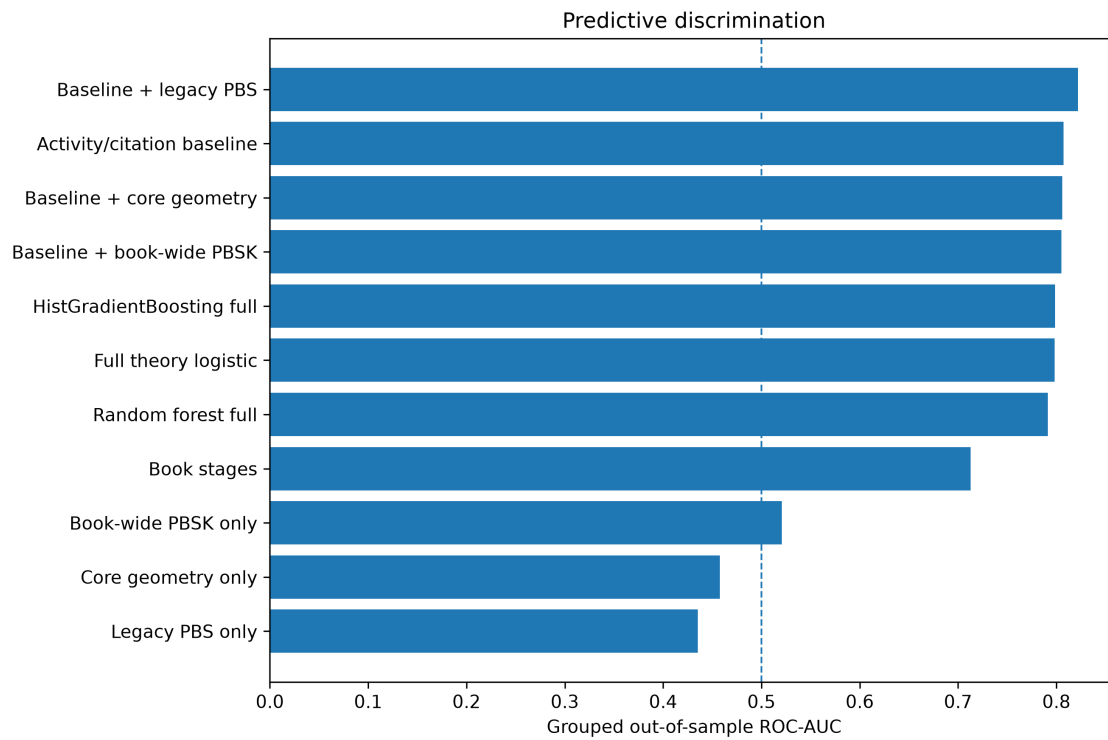


Figure 13: Software-generated figure: figure 14 power validation performance. This figure is reproduced without numerical alteration from the frozen PBSK v4.0.0 run.

Software-generated tables

Supplementary Table S1

Table 2: Software-generated result preview: table_ablation_openalex.csv.

removed component	n	n features	roc auc
none	92	15	0.4997
stage_possibility	92	14	0.5019
stage_question	92	14	0.4964
stage_pattern	92	14	0.4986
stage_imagination	92	14	0.4975
stage_insight	92	14	0.4986
stage_discovery_readiness	92	14	0.5085
bookwide_pbsk	92	14	0.4986
core_geometry_score	92	14	0.5151

Note. Preview shows 9 of 9 rows and 4 of 7 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_ablation_openalex.csv`.

Supplementary Table S2

Table 3: Software-generated result preview: table_book_operationalization.csv.

chapter	part	title	book theme
1	The Invisible World	The Day Before the Idea	Pre-existence of novelty
2	The Invisible World	The Territory of Possibility	Possibility space
3	The Invisible World	The Things That Do Not Yet ...	Unrealized futures
4	The Invisible World	The Invisible Door	Anomalies/openings
5	The Invisible World	The Shape of the Unseen	Hidden structure
6	The Invisible World	The First Impossible Thing	Boundary challenge
7	The Invisible World	The Library of Futures	Future possibility space
8	The Invisible World	Why Reality Is Incomplete	Knowledge incompleteness
9	The Question	The Difference Between Seei...	Attention to weak signals
10	The Question	The Question Nobody Asked	Unasked questions

Note. Preview shows 10 of 40 rows and 4 of 7 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_book_operationalization.csv`.

Supplementary Table S3

Table 4: Software-generated result preview: table_creative_universe_generalization.csv.

model	book realm	n	roc auc
Activity/citation baseline	Life	47	0.5458
Activity/citation baseline	Machines	34	0.428
Activity/citation baseline	Nature	11	1
Baseline + book-wide PBSK	Life	47	0.5646
Baseline + book-wide PBSK	Machines	34	0.3788
Baseline + book-wide PBSK	Nature	11	1
Baseline + core geometry	Life	47	0.4917
Baseline + core geometry	Machines	34	0.3333
Baseline + core geometry	Nature	11	1
Baseline + legacy PBS	Life	47	0.4521

Note. Preview shows 10 of 33 rows and 4 of 6 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_creative_universe_generalization.csv`.

Supplementary Table S4

Table 5: Software-generated result preview: table_cross_source_agreement.csv.

metric	n	pearson r	spearman rho
bookwide_pbsk	10	0.1773	0.05455
core_geometry_score	10	0.4946	0.6121
skg	10	-0.4605	-0.2242
possibility_pressure	10	-0.3113	-0.4667
question_pressure	10	-0.08048	0.06667
collision_index	10	0.1724	0.0303
insight_coherence	10	-0.2096	-0.05628
sequence_order_score	10	-0.5818	-0.6636

Note. Preview shows 8 of 8 rows and 4 of 5 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_cross_source_agreement.csv`.

Supplementary Table S5

Table 6: Software-generated result preview: table_figure_generation_qc.csv.

figure	status	message
figure_01_benchmark_timeline	OK	
figure_02_bookwide_pbsk_tra...	OK	
figure_03_book_stage_profil...	OK	
figure_04_bookwide_pbsk_dis...	OK	
figure_05_sequence_architec...	OK	
figure_06_predictive_perfor...	OK	
figure_07_temporal_prediction	OK	
figure_08_bookwide_ablation	OK	
figure_09_transformation	OK	
figure_10_creative_universe...	OK	

Note. Preview shows 10 of 14 rows and 3 of 3 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_figure_generation_qc.csv`.

Supplementary Table S6

Table 7: Software-generated result preview: table_hypothesis_tests_openalex.csv.

cohort	offset	metric	family
landmark_exploratory	5	skg	confirmatory
landmark_exploratory	5	collision_index	confirmatory
landmark_exploratory	5	sequence_order_score	confirmatory
landmark_exploratory	5	bookwide_pbsk	confirmatory
landmark_exploratory	5	core_geometry_score	confirmatory
landmark_exploratory	5	pbs	exploratory
landmark_exploratory	5	possibility_pressure	exploratory
landmark_exploratory	5	question_pressure	exploratory
landmark_exploratory	5	anomaly_pressure	exploratory
landmark_exploratory	5	doubt_tension	exploratory

Note. Preview shows 10 of 69 rows and 4 of 12 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_hypothesis_tests_openalex.csv`.

Supplementary Table S7

Table 8: Software-generated result preview: table_incremental_value_openalex.csv.

baseline model	augmented model	n	n groups
Activity/citation baseline	Baseline + legacy PBS	92	12
Activity/citation baseline	Baseline + book-wide PBSK	92	12
Activity/citation baseline	Baseline + core geometry	92	12
Activity/citation baseline	Full theory logistic	92	12

Note. Preview shows 4 of 4 rows and 4 of 8 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_incremental_value_openalex.csv`.

Supplementary Table S8

Table 9: Software-generated result preview: table_landmark_benchmark.csv.

case id	label	query	domain
crispr	CRISPR-Cas genome editing	CRISPR Cas9 genome editing	Life Sciences
ips	Induced pluripotent stem cells	induced pluripotent stem ce...	Life Sciences
mrna	mRNA vaccine technology	messenger RNA vaccine mRNA	Life Sciences
		...	
alphafold2	AlphaFold2 protein structur...	protein structure predictio...	Life Sciences
singlecell	Single-cell RNA sequencing	single cell RNA sequencing ...	Life Sciences
rnai	RNA interference	RNA interference double str...	Life Sciences
graphene	Graphene isolation	graphene two dimensional ca...	Physical Sciences
transformer	Transformer architecture	attention transformer neura...	Computer Science
resnet	Deep residual networks	deep residual learning resi...	Computer Science
diffusion	Diffusion generative models	diffusion probabilistic gen...	Computer Science

Note. Preview shows 10 of 12 rows and 4 of 8 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_landmark_benchmark.csv`.

Supplementary Table S9

Table 10: Software-generated result preview: table_matched_controls.csv.

control id	match case id	label	query
crispr_c1	crispr	Zinc-finger nuclease engine...	zinc finger nuclease genome...
crispr_c2	crispr	Meganuclease genome enginee...	meganuclease genome enginee...
ips_c1	ips	Adult stem-cell transdiffer...	adult stem cell transdiffer...
ips_c2	ips	Somatic cell nuclear transfer	somatic cell nuclear transf...
mrna_c1	mrna	DNA vaccine platforms	DNA vaccine platform immuni...
mrna_c2	mrna	Viral-vector vaccine platforms	viral vector vaccine platform
alphafold2_c1	alphafold2	Protein threading methods	protein threading structure...
alphafold2_c2	alphafold2	Homology modelling	protein homology modeling s...
singlecell_c1	singlecell	Bulk RNA sequencing	bulk RNA sequencing transcr...
singlecell_c2	singlecell	Microarray transcriptomics	microarray transcriptome ge...

Note. Preview shows 10 of 24 rows and 4 of 6 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_matched_controls.csv`.

Supplementary Table S10

Table 11: Software-generated result preview: table_oof_predictions_openalex.csv.

region id	match case id	offset	target
crispr	crispr	5	1
crispr	crispr	3	1
crispr	crispr	1	1
ips	ips	5	1
ips	ips	3	1
ips	ips	1	1
mrna	mrna	5	1
mrna	mrna	3	1
mrna	mrna	1	1
alphafold2	alphafold2	1	1

Note. Preview shows 10 of 1012 rows and 4 of 7 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_oof_predictions_openalex.csv`.

Supplementary Table S11

Table 12: Software-generated result preview: table_openalex_bookwide_features.csv.

source	region id	offset	cutoff year
OpenAlex	crispr	5	2007
OpenAlex	crispr	3	2009
OpenAlex	crispr	1	2011
OpenAlex	ips	5	2001
OpenAlex	ips	3	2003
OpenAlex	ips	1	2005
OpenAlex	mrna	5	2015
OpenAlex	mrna	3	2017
OpenAlex	mrna	1	2019
OpenAlex	alphafold2	5	2016

Note. Preview shows 10 of 108 rows and 4 of 50 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_openalex_bookwide_features.csv`.

Supplementary Table S12

Table 13: Software-generated result preview: table_power_validation_ablation.csv.

removed component	n	n features	roc auc
none	1080	15	0.8027
stage_possibility	1080	14	0.79
stage_question	1080	14	0.8075
stage_pattern	1080	14	0.8046
stage_imagination	1080	14	0.8024
stage_insight	1080	14	0.804
stage_discovery_readiness	1080	14	0.8037
bookwide_pbsk	1080	14	0.8063
core_geometry_score	1080	14	0.8039

Note. Preview shows 9 of 9 rows and 4 of 7 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_power_validation_ablation.csv`.

Supplementary Table S13

Table 14: Software-generated result preview: table_power_validation_controls.csv.

topic id	topic name	domain	field
https://openalex.org/T14341	Education, Technology, and ...	Physical Sciences	Environmental Science
https://openalex.org/T12643	Sustainability and Ecologic...	Physical Sciences	Environmental Science
https://openalex.org/T14054	Humic Substances and Bio-Or...	Life Sciences	Agricultural and Biological...
https://openalex.org/T13218	Sesame and Sesamin Research	Life Sciences	Agricultural and Biological...
https://openalex.org/T14027	21st Century Education and ...	Social Sciences	Social Sciences
https://openalex.org/T13982	Social Movements and Cultur...	Social Sciences	Social Sciences
https://openalex.org/T14407	Athletic Training and Educa...	Health Sciences	Health Professions
https://openalex.org/T14447	Trade Secret Protection Met...	Health Sciences	Health Professions
https://openalex.org/T13975	Energy, Economy, and Techno...	Physical Sciences	Energy
https://openalex.org/T14374	Statistical Methods in Epid...	Physical Sciences	Mathematics

Note. Preview shows 10 of 240 rows and 4 of 17 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_power_validation_controls.csv`.

Supplementary Table S14

Table 15: Software-generated result preview: table_power_validation_cross_source_agreement.csv.

metric	n	pearson r	spearman rho
bookwide_pbsk	79	0.09993	-0.04734
core_geometry_score	79	-0.02655	-0.04575
skg	79	0.04281	-0.02018
possibility_pressure	79	0.07772	0.1397
question_pressure	79	0.1109	0.3472
collision_index	79	-0.009221	-0.02438
insight_coherence	79	0.04573	0.004366
sequence_order_score	79	0.168	0.1482

Note. Preview shows 8 of 8 rows and 4 of 5 columns. The complete machine-readable table is supplied in supplementary_data/results_tables/ as table_power_validation_cross_source_agreement.csv.

Supplementary Table S15

Table 16: Software-generated result preview: table_power_validation_features_openalex.csv.

source	region id	offset	cutoff year
OpenAlex	PV_0000_P	5	2014
OpenAlex	PV_0000_P	3	2016
OpenAlex	PV_0000_P	1	2018
OpenAlex	PV_0000_C1	5	2014
OpenAlex	PV_0000_C1	3	2016
OpenAlex	PV_0000_C1	1	2018
OpenAlex	PV_0000_C2	5	2014
OpenAlex	PV_0000_C2	3	2016
OpenAlex	PV_0000_C2	1	2018
OpenAlex	PV_0001_P	5	2014

Note. Preview shows 10 of 1080 rows and 4 of 50 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_power_validation_features_openalex.csv`.

Supplementary Table S16

Table 17: Software-generated result preview: table_power_validation_features_pubmed.csv.

source	region id	offset	cutoff year
PubMed	PV_0001_P	5	2014
PubMed	PV_0001_P	3	2016
PubMed	PV_0001_P	1	2018
PubMed	PV_0001_C1	5	2014
PubMed	PV_0001_C1	3	2016
PubMed	PV_0001_C1	1	2018
PubMed	PV_0001_C2	5	2014
PubMed	PV_0001_C2	3	2016
PubMed	PV_0001_C2	1	2018
PubMed	PV_0003_C1	5	2014

Note. Preview shows 10 of 354 rows and 4 of 50 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_power_validation_features_pubmed.csv`.

Supplementary Table S17

Table 18: Software-generated result preview: table_power_validation_hypotheses.csv.

cohort	offset	metric	family
power_validation_confirmatory	5	skg	confirmatory
power_validation_confirmatory	5	collision_index	confirmatory
power_validation_confirmatory	5	sequence_order_score	confirmatory
power_validation_confirmatory	5	bookwide_pbsk	confirmatory
power_validation_confirmatory	5	core_geometry_score	confirmatory
power_validation_confirmatory	5	pbs	exploratory
power_validation_confirmatory	5	possibility_pressure	exploratory
power_validation_confirmatory	5	question_pressure	exploratory
power_validation_confirmatory	5	anomaly_pressure	exploratory
power_validation_confirmatory	5	doubt_tension	exploratory

Note. Preview shows 10 of 69 rows and 4 of 12 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_power_validation_hypotheses.csv`.

Supplementary Table S18

Table 19: Software-generated result preview: table_power_validation_incremental_value.csv.

baseline model	augmented model	n	n groups
Activity/citation baseline	Baseline + legacy PBS	1080	120
Activity/citation baseline	Baseline + book-wide PBSK	1080	120
Activity/citation baseline	Baseline + core geometry	1080	120
Activity/citation baseline	Full theory logistic	1080	120

Note. Preview shows 4 of 4 rows and 4 of 8 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_power_validation_incremental_value.csv`.

Supplementary Table S19

Table 20: Software-generated result preview: table_power_validation_oof_predictions.csv.

region id	match case id	offset	target
PV_0000_P	PV_0000	5	1
PV_0000_P	PV_0000	3	1
PV_0000_P	PV_0000	1	1
PV_0000_C1	PV_0000	5	0
PV_0000_C1	PV_0000	3	0
PV_0000_C1	PV_0000	1	0
PV_0000_C2	PV_0000	5	0
PV_0000_C2	PV_0000	3	0
PV_0000_C2	PV_0000	1	0
PV_0001_P	PV_0001	5	1

Note. Preview shows 10 of 11880 rows and 4 of 7 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_power_validation_oof_predictions.csv`.

Supplementary Table S20

Table 21: Software-generated result preview: table_power_validation_positive_topics.csv.

topic id	topic name	domain	field
https://openalex.org/T11926	Subterranean biodiversity a...	Physical Sciences	Earth and Planetary Sciences
https://openalex.org/T12710	Engineering and Agricultura...	Life Sciences	Agricultural and Biological...
https://openalex.org/T14443	COVID-19, Geopolitics, Tech...	Social Sciences	Social Sciences
https://openalex.org/T13945	Legal, Health, Environmenta...	Health Sciences	Medicine
https://openalex.org/T13647	AI and Big Data Applications	Physical Sciences	Computer Science
https://openalex.org/T14182	Technology-Enhanced Educati...	Social Sciences	Social Sciences
https://openalex.org/T11396	Artificial Intelligence in ...	Health Sciences	Health Professions
https://openalex.org/T11307	Domain Adaptation and Few-S...	Physical Sciences	Computer Science
https://openalex.org/T11101	Online Learning Methods and...	Social Sciences	Social Sciences
https://openalex.org/T12702	Brain Tumor Detection and C...	Life Sciences	Neuroscience

Note. Preview shows 10 of 120 rows and 4 of 16 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_power_validation_positive_topics.csv`.

Supplementary Table S21

Table 22: Software-generated result preview: table_power_validation_predictive_performance.csv.

model	n	n groups	n features
Activity/citation baseline	1080	120	7
Legacy PBS only	1080	120	1
Book-wide PBSK only	1080	120	1
Core geometry only	1080	120	1
Book stages	1080	120	6
Baseline + legacy PBS	1080	120	14
Baseline + book-wide PBSK	1080	120	8
Baseline + core geometry	1080	120	8
Full theory logistic	1080	120	36
Random forest full	1080	120	36

Note. Preview shows 10 of 11 rows and 4 of 14 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_power_validation_predictive_performance.csv`.

Supplementary Table S22

table_power_validation_pubmed_replication.csv

The frozen run produced an empty table for this analysis. The zero-row CSV is retained in `supplementary_data/results_tables/` for provenance.

Supplementary Table S23

Table 23: Software-generated result preview: table_power_validation_temporal_performance.csv.

offset	model	n	n groups
5	Activity/citation baseline	360	120
5	Book-wide PBSK only	360	120
5	Core geometry only	360	120
5	Baseline + core geometry	360	120
5	Full theory logistic	360	120
3	Activity/citation baseline	360	120
3	Book-wide PBSK only	360	120
3	Core geometry only	360	120
3	Baseline + core geometry	360	120
3	Full theory logistic	360	120

Note. Preview shows 10 of 15 rows and 4 of 15 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_power_validation_temporal_performance.csv`.

Supplementary Table S24

Table 24: Software-generated result preview: table_predictive_performance_openalex.csv.

model	n	n groups	n features
Activity/citation baseline	92	12	7
Legacy PBS only	92	12	1
Book-wide PBSK only	92	12	1
Core geometry only	92	12	1
Book stages	92	12	6
Baseline + legacy PBS	92	12	14
Baseline + book-wide PBSK	92	12	8
Baseline + core geometry	92	12	8
Full theory logistic	92	12	36
Random forest full	92	12	36

Note. Preview shows 10 of 11 rows and 4 of 14 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_predictive_performance_openalex.csv`.

Supplementary Table S25

Table 25: Software-generated result preview: table_pubmed_bookwide_features.csv.

source	region id	offset	cutoff year
PubMed	ips	5	2001
PubMed	ips	3	2003
PubMed	ips	1	2005
PubMed	mrna	5	2015
PubMed	mrna	3	2017
PubMed	mrna	1	2019
PubMed	alphafold2	5	2016
PubMed	alphafold2	3	2018
PubMed	alphafold2	1	2020
PubMed	singlecell	5	2004

Note. Preview shows 10 of 48 rows and 4 of 50 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_pubmed_bookwide_features.csv`.

Supplementary Table S26

table_pubmed_replication_hypotheses.csv

The frozen run produced an empty table for this analysis. The zero-row CSV is retained in `supplementary_data/results_tables/` for provenance.

Supplementary Table S27

Table 26: Software-generated result preview: table_pubmed_replication_transformation.csv.

metric	n pairs	positive mean	control mean
post_growth	6	0.8578	0.7425
topic_diffusion	6	0.5868	0.4704
normalization_index	6	0.7508	0.5793
post_adoption	6	0	0
transformation	6	0.5489	0.448

Note. Preview shows 5 of 5 rows and 4 of 9 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_pubmed_replication_transformation.csv`.

Supplementary Table S28

Table 27: Software-generated result preview: table_region_features_all_sources.csv.

source	region id	offset	cutoff year
OpenAlex	crispr	5	2007
OpenAlex	crispr	3	2009
OpenAlex	crispr	1	2011
OpenAlex	ips	5	2001
OpenAlex	ips	3	2003
OpenAlex	ips	1	2005
OpenAlex	mrna	5	2015
OpenAlex	mrna	3	2017
OpenAlex	mrna	1	2019
OpenAlex	alphafold2	5	2016

Note. Preview shows 10 of 1590 rows and 4 of 50 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_region_features_all_sources.csv`.

Supplementary Table S29

Table 28: Software-generated result preview: table_retrieval_qc.csv.

source	cohort	region id	match case id
OpenAlex	landmark	crispr	crispr
OpenAlex	landmark	crispr	crispr
OpenAlex	landmark	ips	ips
OpenAlex	landmark	ips	ips
OpenAlex	landmark	mrna	mrna
OpenAlex	landmark	mrna	mrna
OpenAlex	landmark	alphafold2	alphafold2
OpenAlex	landmark	alphafold2	alphafold2
OpenAlex	landmark	singlecell	singlecell
OpenAlex	landmark	singlecell	singlecell

Note. Preview shows 10 of 1068 rows and 4 of 12 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_retrieval_qc.csv`.

Supplementary Table S30

Table 29: Software-generated result preview: table_sequence_architecture.csv.

offset	class	n	mean sequence order score
5	breakthrough	8	0.4458
5	control	19	0.4246
3	breakthrough	10	0.3833
3	control	22	0.4076
1	breakthrough	11	0.3879
1	control	22	0.4242

Note. Preview shows 6 of 6 rows and 4 of 5 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_sequence_architecture.csv`.

Supplementary Table S31

Table 30: Software-generated result preview: table_temporal_leakage_audit.csv.

source	region id	offset	cutoff year
OpenAlex	crispr	5	2007
OpenAlex	crispr	3	2009
OpenAlex	crispr	1	2011
OpenAlex	ips	5	2001
OpenAlex	ips	3	2003
OpenAlex	ips	1	2005
OpenAlex	mrna	5	2015
OpenAlex	mrna	3	2017
OpenAlex	mrna	1	2019
OpenAlex	alphafold2	5	2016

Note. Preview shows 10 of 1590 rows and 4 of 6 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_temporal_leakage_audit.csv`.

Supplementary Table S32

Table 31: Software-generated result preview: table_temporal_predictive_performance.csv.

offset	model	n	n groups
5	Activity/citation baseline	27	12
5	Book-wide PBSK only	27	12
5	Core geometry only	27	12
5	Baseline + core geometry	27	12
5	Full theory logistic	27	12
3	Activity/citation baseline	32	12
3	Book-wide PBSK only	32	12
3	Core geometry only	32	12
3	Baseline + core geometry	32	12
3	Full theory logistic	32	12

Note. Preview shows 10 of 15 rows and 4 of 15 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_temporal_predictive_performance.csv`.

Supplementary Table S33

Table 32: Software-generated result preview: table_transformation_outcomes.csv.

source	cohort	region id	match case id
OpenAlex	landmark	crispr	crispr
OpenAlex	landmark	ips	ips
OpenAlex	landmark	mrna	mrna
OpenAlex	landmark	alphafold2	alphafold2
OpenAlex	landmark	singlecell	singlecell
OpenAlex	landmark	rnai	rnai
OpenAlex	landmark	graphene	graphene
OpenAlex	landmark	transformer	transformer
OpenAlex	landmark	resnet	resnet
OpenAlex	landmark	diffusion	diffusion

Note. Preview shows 10 of 414 rows and 4 of 15 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_transformation_outcomes.csv`.

Supplementary Table S34

Table 33: Software-generated result preview: table_transformation_tests_openalex.csv.

metric	n pairs	positive mean	control mean
post_growth	12	0.6401	0.5762
topic_diffusion	12	0.4567	0.4954
normalization_index	12	0.6143	0.5559
post_adoption	12	0.5599	0.564
transformation	12	0.5678	0.5479

Note. Preview shows 5 of 5 rows and 4 of 9 columns. The complete machine-readable table is supplied in `supplementary_data/results_tables/` as `table_transformation_tests_openalex.csv`.

Operationalization and software provenance

The computational formulas and configuration underlying these tables are described in the main manuscript; the book-derived conceptual arc is cited as the motivating source (Akhtar, 2026). The exact frozen software files used for the analysis are included under `supplementary_data/software_snapshot/`, including the feature-engineering, modeling, statistics, power-validation, and configuration files. This snapshot is included to ensure that the numerical tables can be traced to executable definitions rather than prose alone.

References cited in the supplement

Akhtar, M. A. K. (2026). *The impossible idea: How new realities enter the world* [Published under the pen name Shunya]. <https://a.co/d/03buY0qI>